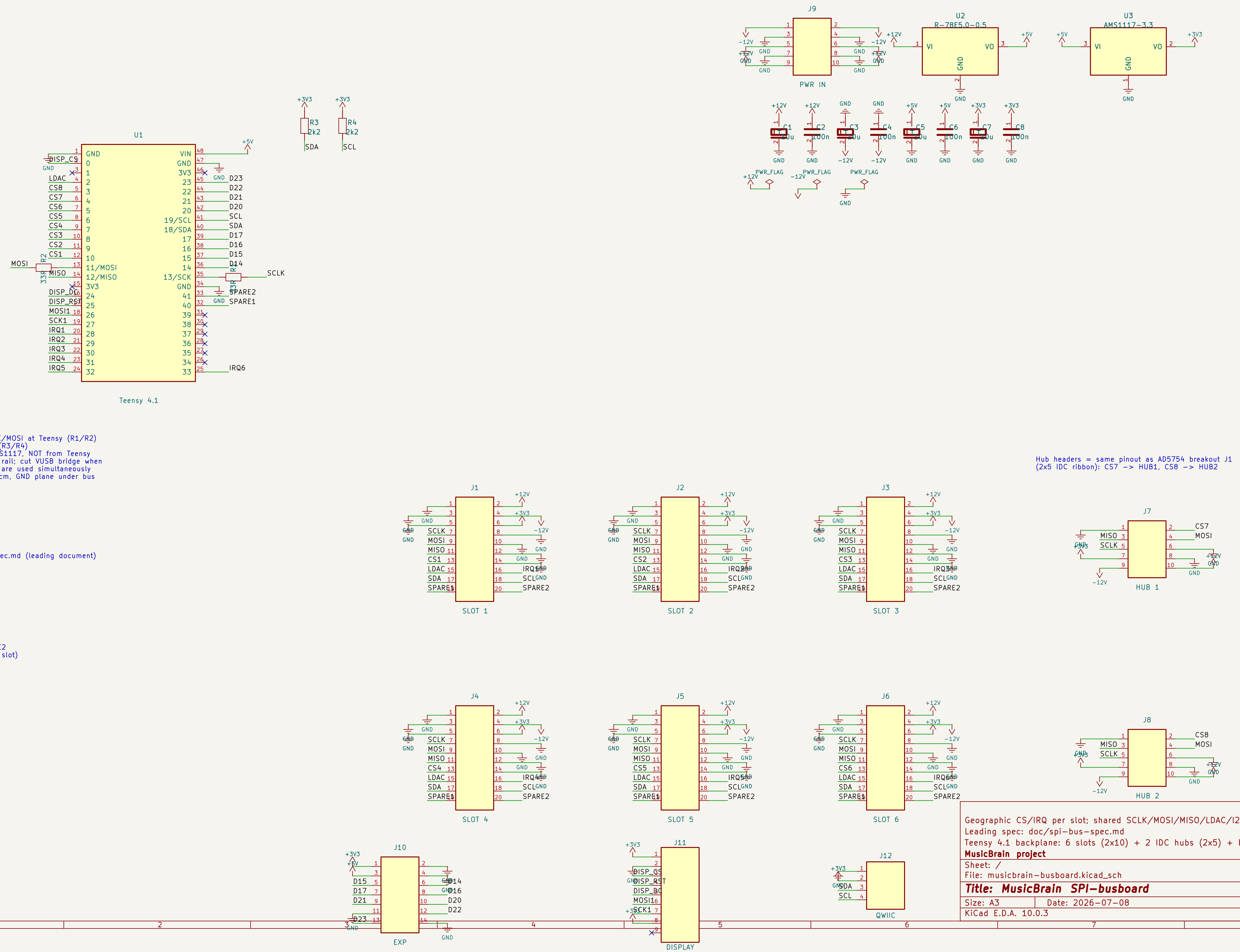
MusicBrain SPI-busboard – Teensy 4.1 backplane



Rules:
- 33R series in SCLK/MOSI at Teensy (R1/R2)
- I2C pull-ups 2k2 (R3/R4)
- +3V3 bus from AMS1117, NOT from Teensy
- Teensy VIN = +5V rail; cut VUSB bridge when USB and bus power are used simultaneously
- backplane <= 20 cm, GND plane under bus

Spec: doc/spi-bus-spec.md (leading document)

Slot pinout (2x10):

1 GND	2 +12V
3 GND	4 -12V
5 GND	6 +3V3
7 SCLK	8 GND
9 MOSI	10 GND
11 MISO	12 GND
13 CS*	14 GND
15 LDAC	16 IRQ*
17 SDA	18 SCL
19 SPARE1	20 SPARE2

(* = geographic per slot)

Hub headers = same pinout as AD5754 breakout J1 (2x5 IDC ribbon): CS7 -> HUB1, CS8 -> HUB2

Geographic CS/IRQ per slot; shared SCLK/MOSI/MISO/LDAC/I2C
Leading spec: doc/spi-bus-spec.md
Teensy 4.1 backplane: 6 slots (2x10) + 2 IDC hubs (2x5) + EXP/DISPLAY/QWIIC
MusicBrain project
Sheet: /
File: musicbrain-busboard.kicad_sch
Title: MusicBrain SPI-busboard
Size: A3 Date: 2026-07-08 **Rev: 1.1**
KiCad E.D.A. 10.0.3 Id: 1/1